

Class - 06

Introduction

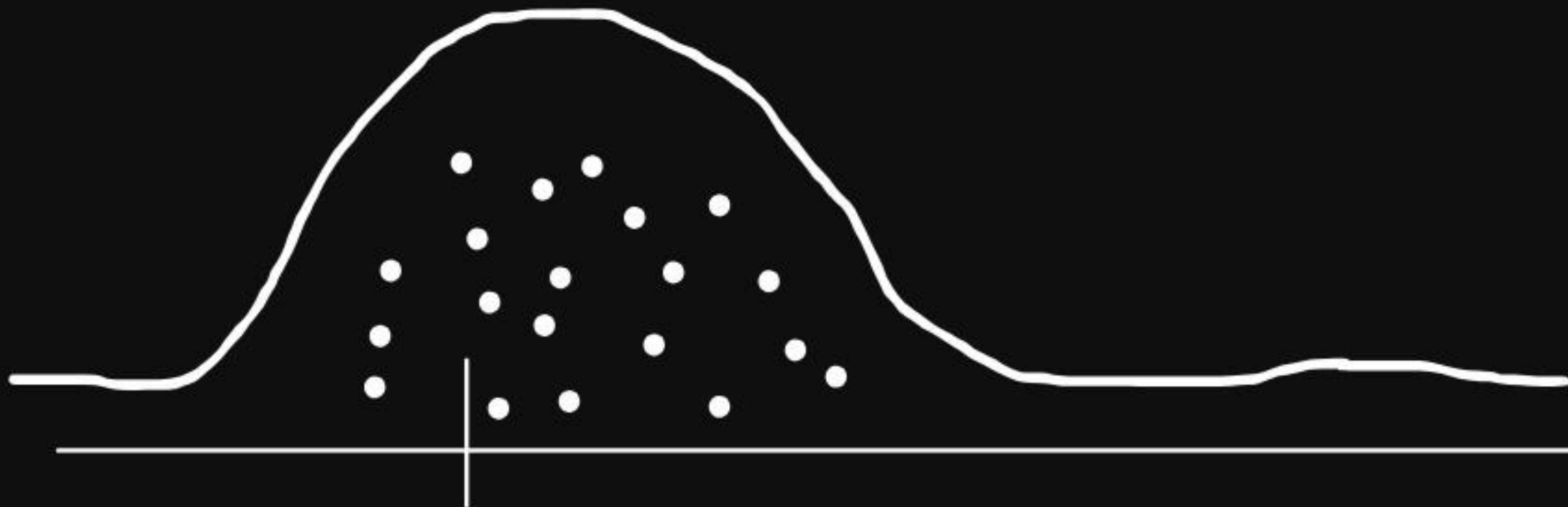
Data Distribution

Data Distribution

First, let's ask " What does data distribution mean?? "

When we collect data (like marks of students, height of people), the values are not randomly scattered, they **follow a Pattern**.

This pattern of how data is **spread out** or **arranged** is called **Data Distribution**.



Data Distribution

Now let's see the simple definition -

" Data distribution tells us how often each value (or range of values) appears in our dataset. "

It shows whether the data is **balanced, skewed, spread out, or concentrated** in one area.

Now let's check some different types of Distribution

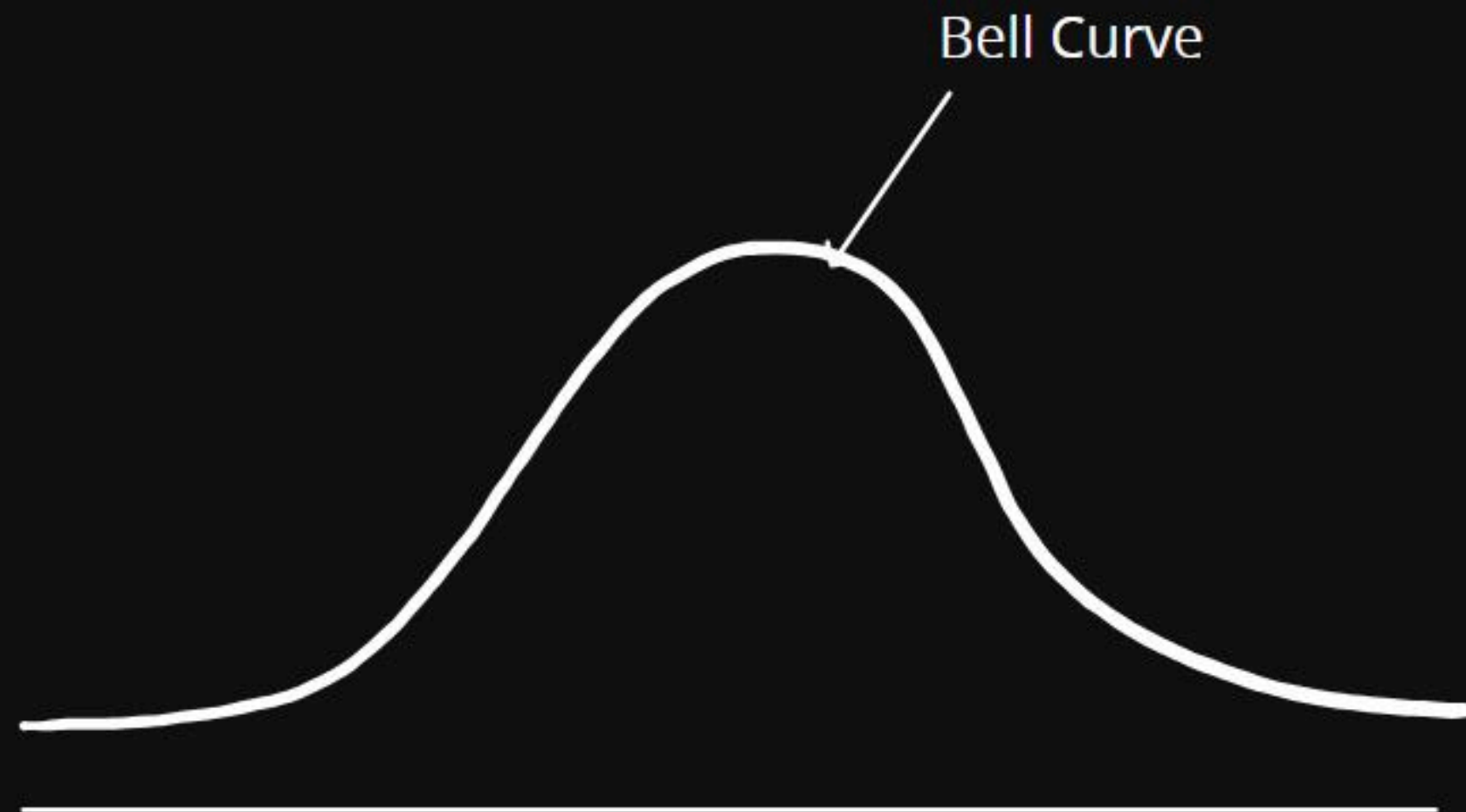
Types of Distribution

Data Distribution

Data Distribution

1. Normal Distribution (Bell-Shaped Curve)

- Most values are **around the average (mean)**.
- Fewer values are very high or very low.
- The shape looks like a **bell curve**.



Data Distribution

2. Skewed Distribution

The skewed means that the data is **not balanced around the mean**.

Instead, most of the data values are **bunched on one side**, and the “Tail ” of the graph stretches longer on the other side.

Now there are 2 types of Skewed Distribution

- Right - Skewed
- Left - Skewed

Data Distribution

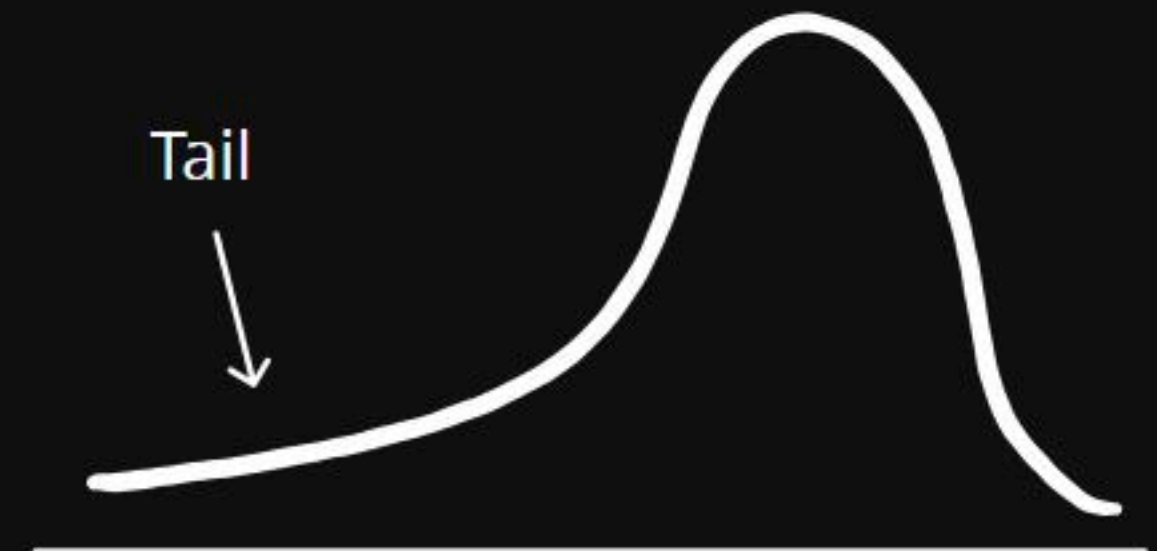
A. Right-Skewed (Positive Skew)

- The **tail is longer on the right side**.
- The values are on the **low side** and few very high.



B. Left - Skewed (Negative Skew)

- The **tail is longer on the left side**.
- The values are on the **high side** and few very low.



Notations

Sample and Population

In Statistics always deals with a **Group of Data**.

But here's the problem , Studying entire group of data is very difficult and can be a headache.

So instead , we usually study on a small group of the large data and then we make conclusion about the whole data.

Small group → **Samples**

Large group → **Population**



Population

- Population is a complete set of Items, People or Measurements.
- It is denoted by **N**.
- The Characteristics of Population is known as **Parameters**.

Samples

- A Sample is a Subset of Population taken for the Study.
- It is denoted by **n**.
- The Characteristics of Sample is known as **Samples**.

MEAN, VARIANCE AND STANDARD DEVIATION

Data Distribution

Mean, Variance and Standard Deviation

1. Mean :-

We have already studied about **Mean**
but here's a quick revision,

- **Mean** is simply the **center** or the **average value** of the data.
- In **Mean** we just Add up all the values, then divide by how many values there are.

The formula of is -

$$\text{Mean} = \frac{\text{Number of values}}{\text{Sum of all values}}$$

Mean, Variance and Standard Deviation

Variance :-

Now let's understand Variance with an example

85, 90, 95, 100, 105

So here the Mean is 95

mean = 95

Now after finding the mean we have to compute the difference from each point and mean and squaring them.

$$(85 - 95)^2 + (90 - 95)^2 + (95 - 95)^2 + (100 - 95)^2 + (105 - 95)^2 = 250$$

Mean, Variance and Standard Deviation

Variance :-

Now just divide the computed difference from the numbers of points in your data

$$\text{Variance} = \frac{250}{5} = 50$$

$$\text{Variance} = 50$$

So the Variance is 50.

Mean, Variance and Standard Deviation

3. Standard Deviation :-

Standard Deviation (SD) is just the **Square Root of Variance**.

It tells us **on average how far each value is from the mean**, in the same units as the data.

The formula of Standard Deviation is -

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

$$\text{Standard Deviation} = \sqrt{\frac{\sum_{i=1}^n (x_i - \text{Mean})^2}{n}}$$

Mean, Variance and Standard Deviation

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Mean, Variance and Standard Deviation

Standard Deviation :-

In the variance we saw an example and in the example we were getting,

$$\text{Variance} = 50$$

$$\text{Standard Deviation} = \sqrt{50} = 7.071$$

So the Standard Deviation is 7.071

Notations

Measure	population	Sample
Mean	μ	$\bar{x}$
Variance	σ^2	s
Standard Deviation	σ	s
Size	N	n